HoTHP: Hyperbolic Rotary Position Embedding-based Transformer Hawkes Process

Length extrapolation in Temporal Point Processes

Hugo Ramos Soares

Universidade Federal do Ceará

Orientador: César Lincoln C. Mattos (UFC)
Coorientador: João Paulo Pordeus Gomes (UFC)

Banca: José Antonio Fernandes de Macêdo (UFC)
Diego Parente Paiva Mesquita (FGV)

Proposta – July 2026

Outline

Real-world event sequences

Many phenomena happen in continuous time and we need to model them:

Telemetry
system failures, infrastructure alerts

Social networks
retweets, likes, reply cascades

Health
admissions, exams, prescriptions of a patient

Maintenance
failures in industrial equipment

What do they have in common?

Each domain produces an ordered sequence of events with timestamps:

$$\mathcal{S} = \{(t_1, k_1), (t_2, k_2), \ldots, (t_N, k_N)\}$$

where t_i is the time and k_i is the type (mark) of event i.

What do we want to model?

Given the past events, we want to answer:

- **When** will the next event happen?
- **Of which type** will it be?

The central tool is the **conditional intensity**:

$$\lambda(t \mid \mathcal{H}_t) = \lim_{\Delta t \to 0} \frac{P(\text{event in } [t, t + \Delta t) \mid \mathcal{H}_t)}{\Delta t}$$

where $\mathcal{H}_t$ is the history of all events before t.

Learning objective

Learn $\lambda(t)$ from data so that the model generalizes to sequences with lengths different from those seen during training.

The practical problem: train short, test long

- Collecting long event sequences is often expensive or impractical.
- We would like to **train on short sequences** and run the model on **arbitrarily longer** ones [5].
- Transformer-based point process models degrade sharply when the test sequence is longer than the training one. The ability to remain stable in this regime is called *length extrapolation*.

Contribution of this work: HoTHP

A Transformer Hawkes Process whose attention kernel is **hyperbolic** instead of trigonometric. It embeds an **exponential recency bias** directly in the attention, mirroring the structure of the Hawkes process, and keeps the likelihood stable under length extrapolation.

Objective and research question

Research question

Does replacing the **oscillatory** kernel with a kernel that has **exponential decay** in the attention mechanism improve the length extrapolation ability of neural temporal point processes?

To answer it, we:

- propose and implement the **HoTHP**, adapting HoPE from discrete positions to continuous time instants;
- evaluate it under *train short, test long* on **synthetic** data (known generating process) and **real** data from varied domains;
- analyze *under which conditions* the recency bias is beneficial, neutral, or harmful (learned intensities, accuracy, RMSE, and a temporal-scale ablation).

The Poisson distribution

The question it answers: how many events will we see in a fixed window, if they happen independently at a constant average rate?

Definition [6]

If the *expected number* of events in the window is $\lambda > 0$, the probability of observing exactly k events is

$$\Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \ldots$$

- X: a **random variable**, the count of events in the window (unknown before we observe it);
- k: a **concrete value** that count may take $(0, 1, 2, \ldots)$;
- $\Pr(X = k)$: the **probability** that the observed count X equals exactly k.

- λ is both the **mean** and the **variance** of the count.
- Approximates a binomial: n **trials**, each succeeding with **small** probability p; expected count $\lambda = np$ (e.g., n characters on a page, each misprinted with probability p).

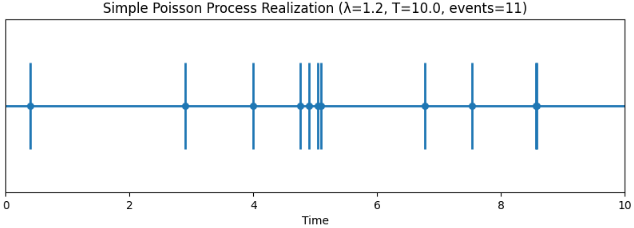
The Poisson process

Now let events keep happening over time, at a constant **rate** λ (events per unit time), and let $N(T)$ count them up to time T [6]. Roughly:

- events in short, **non-overlapping** intervals happen **independently**;
- each short interval gets **at most one** event, with chance proportional to λ.

Then $N(T)$ is Poisson with mean λT (rate times time):

$$\Pr\{N(T) = n\} = \frac{(\lambda T)^n}{n!} e^{-\lambda T}$$



Temporal Point Processes (TPP)

Probabilistic models for sequences of discrete events in **continuous time** [10]. They handle events at irregular, arbitrary times.

- Fully described by the conditional intensity $\lambda(t \mid \mathcal{H}_t)$.
- With **marks** (event types), the sequence is $X = \{(t_1, m_1), \ldots, (t_N, m_N)\}$ and we use K intensities:

$$\lambda_k(t \mid \mathcal{H}_t) = \lim_{\Delta t \to 0} \frac{\Pr(\text{event of type } k \text{ in } [t, t + \Delta t) \mid \mathcal{H}_t)}{\Delta t}$$

Note on the word "mark"

We use *mark* and *type* interchangeably: the mark is a categorical event type, one of K classes.

Loss function: the log-likelihood

Given all events observed in $[0, T]$, TPPs are trained by maximizing the log-likelihood [4]:

$$\mathcal{L} = \sum_{i:\, t_i \le T} \log \lambda_{k_i}(t_i) - \underbrace{\int_0^T \bar{\lambda}(t)\, dt}_{\Lambda}, \qquad \bar{\lambda}(t) = \sum_{k=1}^{K} \lambda_k(t)$$

- First term: rewards a **high intensity** at the moments where events actually occurred (and for the right type k_i).
- Second term (the *compensator* Λ): penalizes intensity mass where no event occurred.
- Training minimizes the **negative** log-likelihood (NLL), our main evaluation metric.

In practice, the integral is estimated numerically, in exactly the same way for every model we compare.

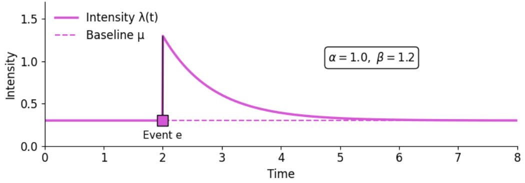
The Hawkes process

Poisson assumes independent events. The **Hawkes process** assumes that **past events raise the probability of future events**, with an influence that decays exponentially [3]:

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \alpha\, e^{-\beta(t - t_i)}$$

- μ: baseline intensity.
- α: instantaneous jump caused by an event.
- β: how fast that influence decays.

The exponential decay is a **recency bias**: recent events matter more.



The Cox process, and where Hawkes sits

Increasingly rich ways of specifying the intensity:

1. **Homogeneous Poisson:** intensity λ is constant.
2. **Non-homogeneous Poisson:** $\lambda(t)$ varies in time, but deterministically.
3. **Cox process** (doubly stochastic Poisson): $\lambda(t)$ is itself a *random* process, driven by factors external to the events.
4. **Hawkes process:** $\lambda(t)$ is random because it depends on the *past events of the process itself*, each one temporarily raising the intensity.

Why this matters here

Hawkes also has a stochastic intensity, but unlike Cox its randomness is **endogenous**: $\lambda(t)$ is a functional of the process's own past (a *self-exciting* process, not a doubly stochastic one). It is exactly this dependence on history, with exponential decay, that motivates the recency bias of our model.

Neural Hawkes Process (NHP)

The classical Hawkes process is limited: influence is always excitatory, constant per type, and decays exponentially. Mei and Eisner [4] relax these assumptions with neural networks.

- Allow $\alpha, \mu \in \mathbb{R}$, so events can **inhibit** as well as excite; a **softplus** keeps the intensity positive.
- Replace the additive sum by a **continuous-time LSTM** (CT-LSTM): the memory cell decays exponentially between events.

$$\lambda_k(t) = f_k\left(\mathbf{w}_k^\top \mathbf{h}(t)\right), \qquad \mathbf{c}(t) = \bar{\mathbf{c}} + (\mathbf{c} - \bar{\mathbf{c}})\, e^{-\delta(t-t_i)}$$

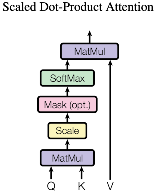
A simple but strong model: still a top benchmark today, and an important baseline in our experiments.

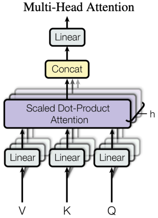
Transformers and the attention mechanism

RNNs (including the CT-LSTM) process events **sequentially**, which is slow and struggles with long sequences. The Transformer [8] removes recurrence: all events are processed at once through **attention**.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- Each event attends to every other event; the weights measure relevance.
- **Multi-head** attention runs h projections in parallel and concatenates them.

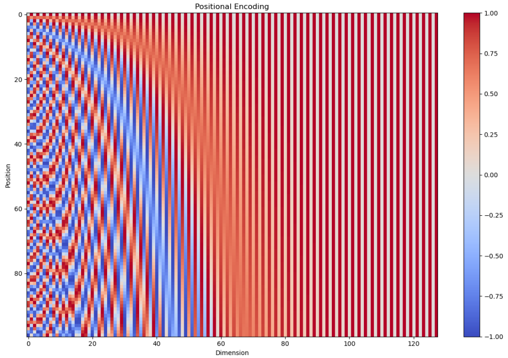




Positional encoding: attention has no order

Attention is **permutation equivariant**: reordering the inputs merely reorders the outputs, so on its own it does not know the order or timing of events. Position must be injected.

- **Original Transformer:** add a sinusoidal encoding of the *discrete position* to each token embedding.
- Each dimension is a sine/cosine of a different frequency: fast on the left, slow on the right, giving every position a unique signature.
- **For TPPs** we need the actual *timestamp*, not just the order, since the moment of the event matters.



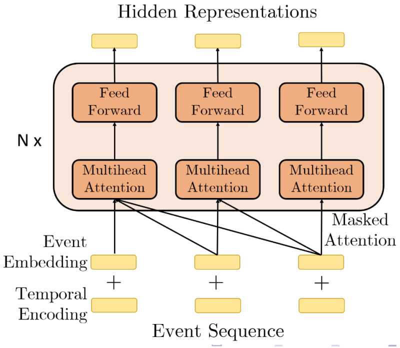
Sinusoidal encoding: position (rows) vs. dimension (columns).

Zuo et al. [10] adapt the Transformer to TPPs with a **temporal encoding** that uses the timestamp t_j instead of a discrete position:

$$[z(t_j)]_i = \begin{cases} \cos\left(t_j/10000^{(i-1)/M}\right), & i \text{ odd}, \\ \sin\left(t_j/10000^{i/M}\right), & i \text{ even}. \end{cases}$$

The encoding is added to the type embedding, then passed through self-attention and a feed-forward network. The intensity is

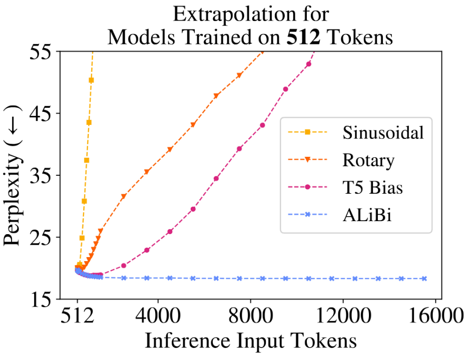
$$\lambda_k(t) = f_k\Big(\underbrace{\alpha_k \frac{t-t_j}{t_j}}_{\text{current}} + \underbrace{\mathbf{w}_k^\top \mathbf{h}(t_j)}_{\text{history}} + \underbrace{b_k}_{\text{base}}\Big).$$



Length extrapolation in language models

A model trained on sequences of length L_{train} should stay stable when tested on $L_{\text{test}} > L_{\text{train}}$ [5].

- Absolute sinusoidal positions produce encoding patterns never seen in training for larger indices: perplexity explodes.
- Rotary encodings (RoPE) help by using *relative* positions, but Press et al. [5] showed they still fail for much longer sequences.
- Only **ALiBi**, with a linear distance penalty, stays flat far beyond the training length.



The same problem in TPPs: train short, test long

We adapt the *train short, test long* protocol [5] to point processes, using the **number of events** as the length.

- Train on sequences of a fixed length (e.g. $L = 50$ events).
- Test on sequences α times longer, $\alpha \in \{1, 2, 5, 10\}$.
- At $\alpha = 10\times$: train on 50 events, test on 500.

What we measure

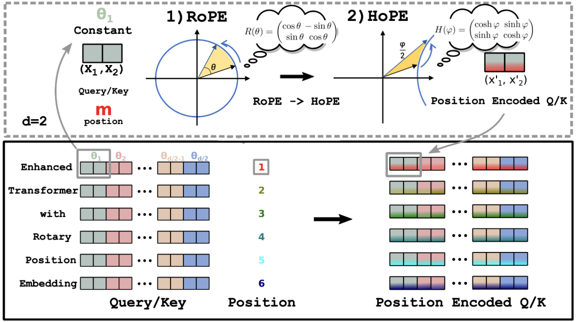
Whether the NLL stays stable as α grows. A model with a proper recency bias should barely degrade.

Rotary Position Embedding (RoPE)

Su et al. [7] noticed that attention only needs the **relative** position. RoPE rotates pairs of embedding dimensions by an angle proportional to the position:

$$\mathbf{R}_j(\Delta) = \begin{bmatrix} \cos(\theta_j\Delta) & \sin(\theta_j\Delta) \\ -\sin(\theta_j\Delta) & \cos(\theta_j\Delta) \end{bmatrix}, \qquad \theta_j = 10000^{-2(j-1)/d}.$$

- Applied to **q** and **k** before the dot product.
- The score ends up depending only on the difference of positions.



RoTHP: RoPE for point processes

Dai et al. [2] bring RoPE to TPPs: rotate by the **timestamp** instead of the discrete index.

- The attention score depends only on $\Delta t = t_i - t_j$.
- This makes the likelihood **time-invariant**: shifting all times by a constant σ leaves the result unchanged.
- Robust to timestamp noise (clock desynchronization), common in real data. New state of the art on several TPP benchmarks.

But there is a catch

The rotary kernel is built from cos and sin: it **oscillates**. This is the limitation our method addresses.

ALiBi and the recency inductive bias

Press et al. [5] take a different route: no embedding at all, just a fixed **linear penalty** on the attention score:

$$\text{softmax}\Big(\mathbf{q}_i\mathbf{K}^\top + m \cdot [-(i-1), \ldots, -2, -1, 0]\Big)$$

- The penalty grows with distance, so distant tokens are suppressed. The slope m is fixed per head, never learned.
- This is an **inductive bias toward recency**, built into the architecture rather than learned from data.

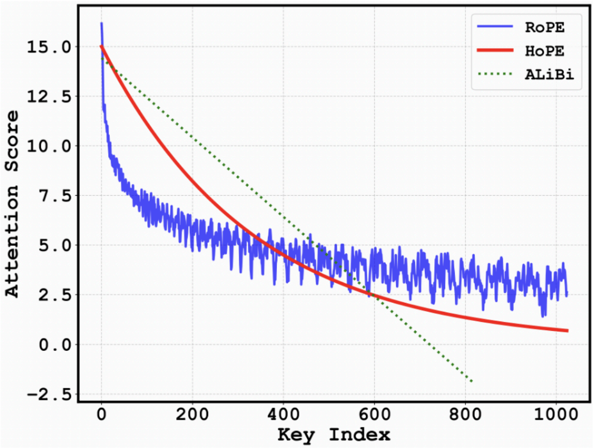
The connection to Hawkes

This is exactly the Hawkes kernel idea (influence $\propto e^{-\beta\Delta t}$ decays with distance). THP and RoTHP dropped this structural guarantee when they replaced the kernel with free attention. **HoTHP puts it back.**

The oscillatory limitation of RoTHP

RoTHP's score depends only on $\Delta t = t_i - t_j$, but its kernel is built from cos and sin, so it **oscillates** instead of decaying.

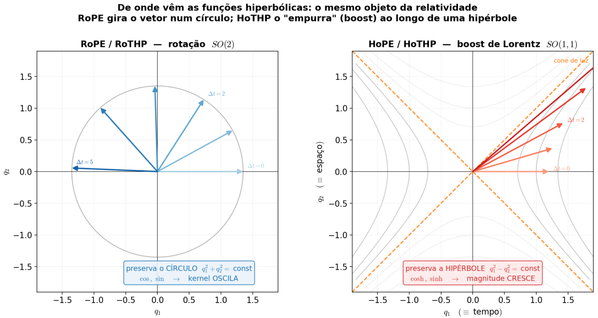
- **RoPE** (blue) decays only *on average*: noisy, non-monotonic, and it never dies out.
- A Hawkes kernel $\varphi(\Delta t) = \alpha e^{-\beta \Delta t}$ is *strictly* decreasing. RoPE violates this.
- In extrapolation the phase may land on a **peak**: a distant event outscores a nearby one.



From the trigonometric kernel to the hyperbolic kernel

Following HoPE [1], replace the rotation block $\mathbf{R}_j$ by a **hyperbolic boost** $\mathbf{B}_j$:

$$\mathbf{R}_j(\Delta t) = \begin{bmatrix} \cos & \sin \\ -\sin & \cos \end{bmatrix} \longrightarrow \mathbf{B}_j(\Delta t) = \begin{bmatrix} \cosh(\theta_j \Delta t) & \sinh(\theta_j \Delta t) \\ \sinh(\theta_j \Delta t) & \cosh(\theta_j \Delta t) \end{bmatrix}$$



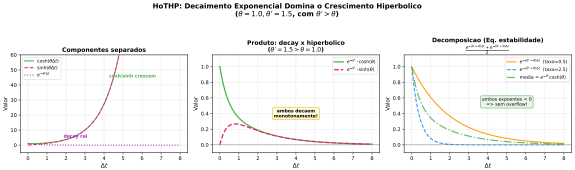
The rotation preserves the **circle** ($\cos, \sin$ oscillate); the Lorentz boost preserves the **hyperbola** ($\cosh, \sinh$ are monotonic). We control this growth with the decay envelope $e^{-|\Delta t|\theta'}$.

Exponential decay and the parameter θ'

HoPE [1] adds a learnable θ' and damps $\mathbf{q}, \mathbf{k}$ so the score decays. In continuous time:

$$g(\Delta t) = e^{-|\Delta t|\,\theta'}\, \mathbf{q}^\top \mathbf{B}(\theta, \Delta t)\, \mathbf{k}$$

- If $\theta' > \max_j \theta_j$, the decay dominates the hyperbolic growth: the score **decays exponentially to zero** with $|\Delta t|$, without oscillation.
- This is a **structural** recency bias: it holds even for Δt never seen in training. This is what enables extrapolation.



HoTHP score: one dimension-pair at a time

Each head splits into $d/2$ **pairs**. Pair p has frequency θ_p and components $(q_1^{(p)}, q_2^{(p)})$ and $(k_1^{(p)}, k_2^{(p)})$.

Score of one pair = hyperbolic rotation by Δt

$$s^{(p)} = (q_1 k_1 + q_2 k_2)\cosh(\theta_p \Delta t) + (q_1 k_2 + q_2 k_1)\sinh(\theta_p \Delta t)$$

Full score = sum over pairs, with the recency envelope $e^{-|\Delta t|\theta'}$ and the usual $1/\sqrt{d}$ scaling ($\Delta t = t_i - t_j$):

$$s_{ij} = \frac{1}{\sqrt{d}} \sum_{p=1}^{d/2} e^{-|\Delta t|\theta'} \, s^{(p)}.$$

We never evaluate cosh / sinh, only exponentials

Step 1. Replace each by its definition (cosh is even, so $\theta_p \Delta t \to \theta_p |\Delta t|$):

$$\cosh(\theta_p \Delta t) = \tfrac{1}{2}\left(e^{\theta_p|\Delta t|} + e^{-\theta_p|\Delta t|}\right), \qquad \sinh(\theta_p \Delta t) = \tfrac{\mathrm{sign}(\Delta t)}{2}\left(e^{\theta_p|\Delta t|} - e^{-\theta_p|\Delta t|}\right).$$

Step 2. Multiply the cosh term by the envelope and *merge the exponents* (this is the whole trick):

$$\begin{aligned} e^{-|\Delta t|\theta'} \cosh(\theta_p \Delta t) &= \tfrac{1}{2}\left(e^{-|\Delta t|\theta'}\, e^{+\theta_p|\Delta t|} + e^{-|\Delta t|\theta'}\, e^{-\theta_p|\Delta t|}\right) \\ &= \tfrac{1}{2}\left(e^{-|\Delta t|(\theta' - \theta_p)} + e^{-|\Delta t|(\theta' + \theta_p)}\right). \end{aligned}$$

Step 3. Since $\theta' > \max_p \theta_p$ by construction, both $\theta' - \theta_p > 0$ and $\theta' + \theta_p > 0$: every exponent is negative, so each term lands in $(0, 1]$, no overflow, no `NaN`. (The sinh term merges the same way, giving $\frac{\mathrm{sign}(\Delta t)}{2}\left(e^{-|\Delta t|(\theta' - \theta_p)} - e^{-|\Delta t|(\theta' + \theta_p)}\right)$.)

Architecture: where the exponential bias acts

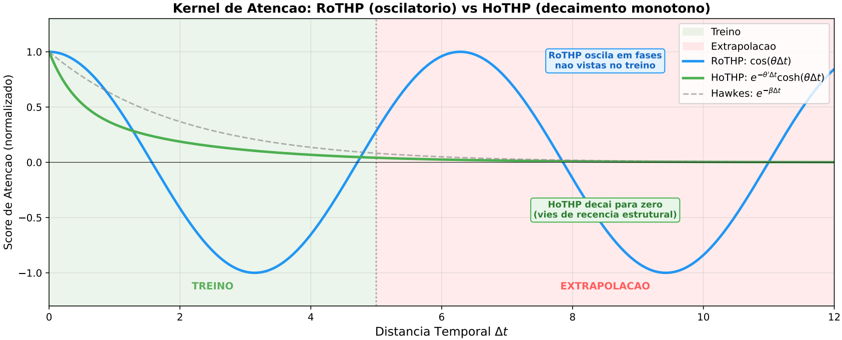
HoTHP is a Transformer encoder with hyperbolic multi-head attention. **No** time information is added to the embedding; time enters only through the kernel. The bias acts in **two distinct steps**:

1. **Hyperbolic attention** $\to \mathbf{h}(t_j)$: the decay decides *which past events* shape the hidden state. Recent events get more weight.
2. **Intensity** $\lambda_k(t) = \mathrm{softplus}(\alpha_k(t - t_j) + \mathbf{w}_k^\top \mathbf{h}(t_j) + b_k)$: linear in the *elapsed* time $t - t_j$, so shifting all timestamps changes nothing; α_k is free (can rise or fall after an event).

Consequence

The recency bias does **not** require the process to be self-exciting. It only requires that recent events be more informative. It helps on self-exciting *and* self-inhibiting processes.

RoTHP vs. HoTHP kernel: training and extrapolation zones



In the **training zone**, both models fit the data. In the **extrapolation zone**, the RoTHP trigonometric kernel keeps oscillating in unseen phases, while the HoTHP hyperbolic kernel decays exponentially to zero, without oscillation, like the classical Hawkes kernel (dashed).

Experimental setup

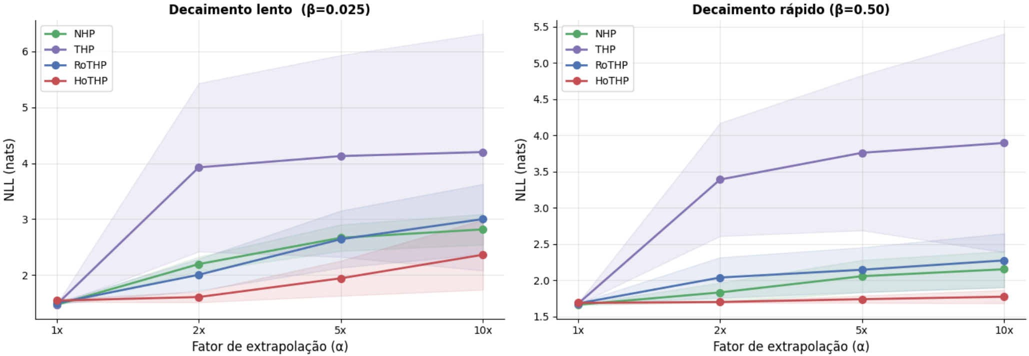
- **Synthetic data:** multivariate Hawkes with exponential kernel, two regimes: *slow decay* (β small) and *fast decay* (β large). We know the ground truth.
- **Real data (EasyTPP [9]):** Taxi, StackOverflow, Amazon, Retweet.
- **Protocol:** train short, test long, factors $\alpha \in \{1, 2, 5, 10\}$ (synthetic) and $\{1, 2, 5\}$ (real).
- **Models:** NHP, THP, RoTHP, HoTHP. THP/RoTHP/HoTHP share the same architecture and intensity, differing **only in how time enters the attention** (a controlled comparison).
- **Metrics:** NLL (main), RMSE, type accuracy. Synthetic: 10 seeds + Wilcoxon signed-rank test. Real: 3 seeds (no Wilcoxon).

Synthetic results: NLL under extrapolation

Scenario	Model	1×	2×	5×	10×
Slow decay	NHP	**1.470**	2.197	2.667	<u>2.817</u>
	THP	<u>1.475</u>	3.926	4.127	4.199
	RoTHP	1.499	<u>2.007</u>	<u>2.642</u>	3.001
	HoTHP	1.545	**1.609**	**1.943**	**2.364**
Fast decay	NHP	**1.663**	<u>1.833</u>	<u>2.057</u>	<u>2.153</u>
	THP	<u>1.677</u>	3.391	3.761	3.897
	RoTHP	1.681	2.039	2.146	2.275
	HoTHP	1.689	**1.701**	**1.740**	**1.774**

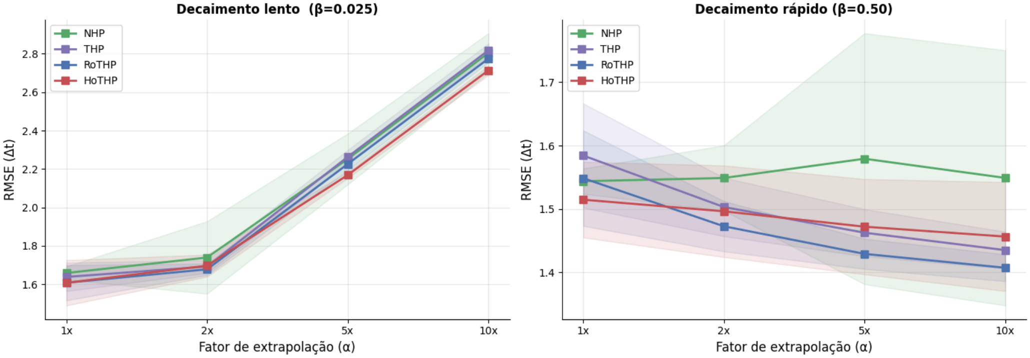
NLL (mean, 10 seeds), lower is better. **HoTHP wins at every extrapolation factor** (2×, 5×, 10×) in both scenarios; the small cost at 1× reverses immediately. Advantage significant (Wilcoxon, $p < 0.05$) in all cases.

Synthetic results: NLL vs. extrapolation factor



In fast decay, the HoTHP NLL grows only 0.085 from $1\times$ to $10\times$ (almost flat), while the THP diverges and RoTHP/NHP degrade more. The recency bias is aligned with the exponential kernel that generated the data.

Synthetic results: RMSE



Unlike the NLL, the RMSE does not separate the models: they stay close at every factor. The HoTHP advantage shows up in the **likelihood** (modeling the full temporal distribution), not in the pointwise prediction of the next interval.

Real data: datasets and NLL

Dataset	Types	L_{train}
Amazon	16	18
StackOverflow	22	20
Taxi	10	7
Retweet	3	50

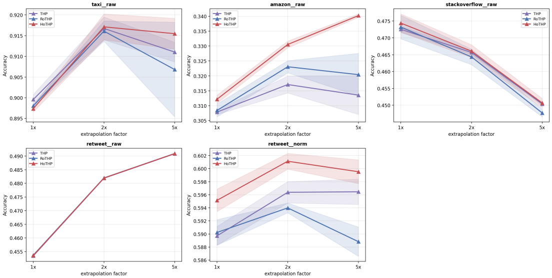
Raw timestamps (shift to zero only), 3 seeds. Retweet is studied separately (ablation).

Dataset	Model	1×	2×	5×
Taxi	THP	−0.273	−0.314	−0.243
	RoTHP	−0.271	−0.317	−0.137
	HoTHP	<u>−0.278</u>	<u>−0.328</u>	<u>−0.266</u>
	NHP	**−0.449**	**−0.506**	**−0.460**
Amazon	THP	2.264	2.237	2.254
	RoTHP	2.268	2.230	2.247
	HoTHP	<u>2.263</u>	<u>2.207</u>	<u>2.187</u>
	NHP	**0.658**	**0.630**	**0.638**
StackOv.	THP	<u>2.744</u>	<u>2.642</u>	**2.519**
	RoTHP	2.766	2.655	2.545
	HoTHP	2.758	2.649	2.536
	NHP	**2.742**	**2.633**	<u>2.524</u>

Within the Transformer family, HoTHP has the best NLL on Amazon and Taxi at every factor, and is the only one that keeps *improving* on Amazon as the sequence grows.

Real data: type-prediction accuracy

Dataset	Model	1×	2×	5×
Amazon	NHP	.293	.306	.315
	THP	.308	.317	.314
	RoTHP	.308	.323	.320
	HoTHP	**.312**	**.331**	**.340**
StackOv.	NHP	.469	.462	.445
	THP	.473	.466	.450
	RoTHP	.473	.464	.448
	HoTHP	**.474**	**.466**	**.451**
Retweet†	NHP	—	—	—
	THP	.590	.596	.597
	RoTHP	.590	.594	.589
	HoTHP	**.595**	**.601**	**.600**



† Retweet normalized (raw timestamps overflow all Transformers); NHP not run there.

HoTHP has the **best accuracy at every factor, on every dataset where recency helps**. It beats the Transformer family and also NHP, whose lead was confined to NLL. On Amazon the margin *grows* with the sequence length (+0.020 over the best baseline at 5×).

What the models learn: conditional intensities

Learned $\lambda^*(t)$ on representative test sequences:

- **Amazon:** intensity grows between events and falls at each event (*self-inhibiting*). Yet HoTHP wins: the most recent event is still the most informative.
- **StackOverflow:** intensity is almost flat (little temporal memory); the models tie.
- **Retweet:** models react only to the initial burst, then collapse to $\lambda^* \approx 0$ (extreme temporal scale).

Confirms the two-step view

The recency bias acts on **attention** (which past events matter), not on the intensity sign. That is why it helps even on a self-inhibiting process.

Ablation on Retweet, and computational cost

Retweet: scale or bias?

Version	Model	1×	5×
Raw	THP	15.9	2.3×10^4
	RoTHP	15.9	7.0×10^5
	HoTHP	15.9	1.4×10^5
Norm.	THP	**−2.180**	<u>0.039</u>
	RoTHP	−2.133	47.6
	HoTHP	<u>−2.139</u>	**−0.651**

The problem was the **scale**. After normalization, HoTHP is the only model with a negative (stable) NLL at 5×.

Cost vs. benefit

- HoTHP computes the kernel per event *pair*: $O(L^2)$.
- On real data it is 1.07× (Taxi) to 2.44× (Retweet) slower than RoTHP, but **cheaper than the NHP** (the strongest NLL baseline).
- Cost is paid once, at training. At inference, what matters is a *valid* result: RoTHP at 5× gives NLL 47.6, HoTHP gives −0.651.

Contributions

- **HoTHP:** a Transformer Hawkes Process with a **hyperbolic** rotary kernel, combining time-invariance with a structural exponential recency bias in a single attention mechanism.
- A learnable θ', parametrized so that $\theta' > \max_j \theta_j$ **by construction**, guaranteeing exponential decay of the score and numerical stability.
- Under *train short, test long*, HoTHP keeps the NLL stable up to $10\times$ on synthetic data (significant by Wilcoxon), and is the best of the Transformer family on Amazon and Taxi.
- An analysis separating *where* the bias acts (attention vs. intensity), explaining why it helps even on self-inhibiting processes.
- Implementation and experiments in the EasyTPP framework.

Timeline to the defense

Qualification: **10 July 2026**. Final defense planned for **August 2026**.

Activity	Jun 2nd	Jul 1st	Jul 2nd	Aug 1st	Aug 2nd
Prepare qualification talk	X	X			
Defense (Jul 10)		X			
Incorporate committee feedback			X	X	
Additional experiments	X		X	X	
Finish writing (conclusion, abstract)	X		X	X	X
Final review with advisor					X
Prepare and present final defense					X

Main items: committee feedback, more seeds for real-data significance tests, refined normalization ablations, and finishing the writing.

Limitations and future work

Limitations

- The exponential bias hurts when distant events are as relevant as recent ones (long-range memory).
- Extreme temporal scales need normalization to avoid numerical saturation (as on raw Retweet).
- θ' is a **single global** decay rate; processes with multiple temporal scales may require multiple rates.

Future work

- More seeds on real data to enable the Wilcoxon test there.
- Replicate the original THP benchmark (where THP outperforms NHP) and apply HoTHP on top of that implementation, to check whether the NHP advantage observed here comes from the EasyTPP reimplementation.
- Multiple decay rates (per head or per dimension pair).

Thank you

Hugo Ramos Soares
Universidade Federal do Ceará
Orientador: César Lincoln C. Mattos
Coorientador: João Paulo Pordeus

References I

[1] Chang Dai, Hongyu Shan, Mingyang Song, and Di Liang.
Hope: Hyperbolic rotary positional encoding for stable long-range dependency modeling in large language models.
2025.

[2] Shan Dai, Anningzhe Gao, Zhuo Li, and Yuhao Du.
Rotary position embedding-based transformer hawkes process for event-type big data.
Big Data Mining and Analytics, 9(1):23–38, 2026.

[3] Alan G. Hawkes.
Spectra of some self-exciting and mutually exciting point processes.
Biometrika, 58(1):83–90, 1971.

[4] Hongyuan Mei and Jason Eisner.
The neural Hawkes process: A neurally self-modulating multivariate point process.
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[5] Ofir Press, Noah A. Smith, and Mike Lewis.
Train short, test long: Attention with linear biases enables input length extrapolation.
In *International Conference on Learning Representations (ICLR)*, 2022.

References II

[6] Sheldon M. Ross.
A First Course in Probability.
Pearson, Boston, MA, 10th edition, 2018.

[7] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu.
Roformer: Enhanced transformer with rotary position embedding.
Neurocomputing, 568, 2024.

[8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin.
Attention is all you need.
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[9] Siqiao Xue, Xiaoming Shi, Zhixuan Chu, Yan Wang, Hongyan Hao, Fan Zhou, Caigao Jiang, Chen Pan, James Y. Zhang, Qingsong Wen, Jun Zhou, and Hongyuan Mei.
EasyTPP: Towards open benchmarking temporal point processes.
In *International Conference on Learning Representations (ICLR)*, 2024.

[10] Simiao Zuo, Haoming Jiang, Zichong Li, Tuo Zhao, and Hongyuan Zha.
Transformer hawkes process.
In *International Conference on Machine Learning (ICML)*, 2020.